HOMEWORK 1: ALGEBRA REVIEW
MATH 161

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Name: _____

Know how to do these problems without the aid of a book or notes.

1. Find the solutions to each polynomial.

   (a) $f(x) = x^2 - 8x + 12$

   $(x-6)(x-2) = x^2 - 2x - 6x - 12 = x^2 - 8x - 12 = 0$

   $\Rightarrow x = 6$ or $x = 2$

   (b) $f(x) = 2x^2 - 9x - 5$

   $(2x+1)(x-5) = 2x^2 - 10x + x - 5 = 2x^2 - 9x - 5 = 0$

   $\Rightarrow 2x+1 = 0$ or $x - 5 = 0$

   $\Rightarrow x = -1/2$ or $x = 5$.

   (c) $f(x) = x^2 - 1$

   $x^2 - 1 = (x+1)(x-1) = 0$

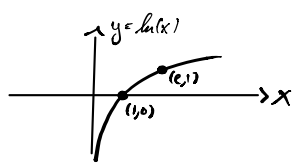
   $\Rightarrow x = -1$ or $x = 1$

   (d) $f(x) = x^2 - 2$

   $x^2 - 2 = 0 \Rightarrow x^2 = 2 \Rightarrow x = \pm\sqrt{2}$

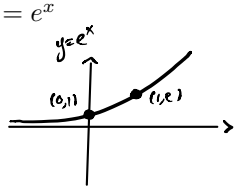
2. Sketch a graph of the following:

   (a) $y = \ln(x)$

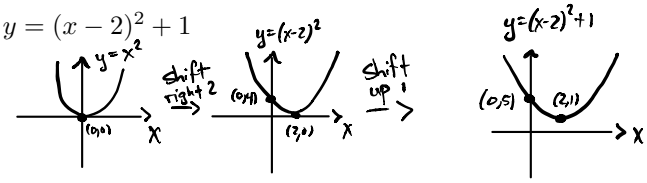


Date: August 27, 2018.

(b) $y = e^x$



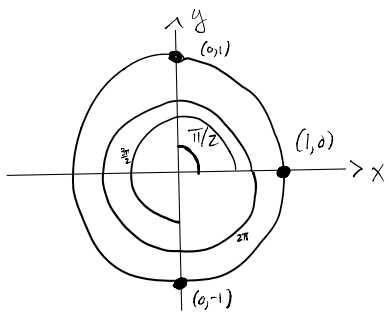
(c) $y = (x-2)^2 + 1$



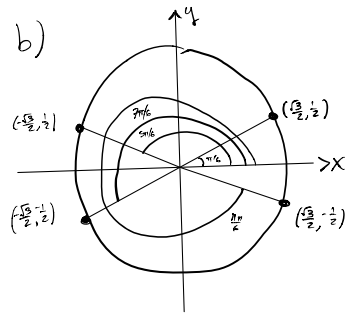
3. Draw the unit circle and fill in the following angles with their corresponding coordinates:

(a) $\pi/2, \pi, 3\pi/2, 2\pi$ **(b)** $\pi/6, 5\pi/6, 7\pi/6, 11\pi/6$ **(c)** $\pi/4, 3\pi/4, 5\pi/4, 7\pi/4$ **(d)** $\pi/3, 2\pi/3, 4\pi/3, 5\pi/3$

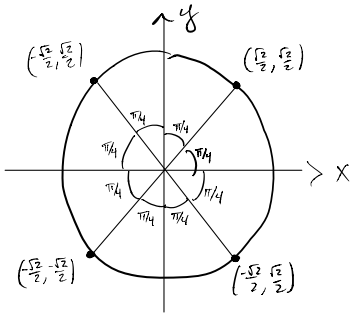
a)



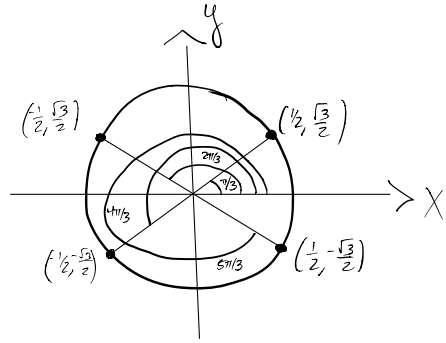
b)



c)



d)



4. Simplify the following expressions:

(a) $x^2(4(x-2)^3) + 2x(x-4)^4$

$4x^2(x-2)^3 = 4x^2\left(x^3 + 3(x^2)(-2) + 3(x)(-2)^2 + (-2)^3\right) = 4x^2(x^3 - 6x^2 + 12x - 8) = 4x^5 - 24x^4 + 48x^3 - 32x^2$

$+ 2x(x-4)^4 = 2x\left(x^4 + 4(x^3)(-4) + 6(x^2)(-4)^2 + 4(x)(-4)^3 + (-4)^4\right) = 2x(x^4 - 16x^3 + 96x^2 - 256x + 256) = 2x^5 - 32x^4 + 192x^3 - 512x^2 + 512x$

$\boxed{6x^5 - 56x^4 + 240x^3 - 544x^2 + 512x}$

(b) $\dfrac{(x^2+3)^2(6) - 6x(2)(x^2+3)(2x)}{(x^2+3)^4}$

$= \dfrac{6(x^2+3)^2 - 24x^2(x^2+3)}{(x^2+3)^4} = \dfrac{(x^2+3)(6(x^2+3) - 24x^2)}{(x^2+3)^4} = \dfrac{6x^2+18-24x^2}{(x^2+3)^3} = \dfrac{-18x^2+18}{(x^2+3)^3}.$

(c) $\dfrac{\frac{1}{x^2} - \frac{1}{9}}{x-3} = \dfrac{1}{x-3}\left(\dfrac{9-x^2}{9x^2}\right) = \dfrac{(3-x)(3+x)}{(x-3)(9x^2)} = \dfrac{-(x-3)(3+x)}{(x-3)(9x^2)} = \dfrac{-3-x}{9x^2}$

(d) $\dfrac{\sqrt{25+x^2} - x(1/2)(25+x^2)^{-1/2}(2x)}{25+x^2} = \dfrac{1}{25+x^2}\left(\sqrt{25+x^2} - \dfrac{x^2}{\sqrt{25+x^2}}\right) = \dfrac{1}{25+x^2}\left(\dfrac{25+x^2}{\sqrt{25+x^2}} - \dfrac{x^2}{\sqrt{25+x^2}}\right)$

$= \dfrac{1}{25+x^2}\left(\dfrac{25}{\sqrt{25+x^2}}\right) = \dfrac{25}{(25+x^2)^{3/2}}$

5. Solve the following inequalities:

(a) $\frac{x}{2} - 1 < 3x + 9$

$\Rightarrow -1-9 = -10 < 3x - \frac{x}{2} = \frac{6x}{2} - \frac{x}{2} = \frac{5x}{2}$

$\Rightarrow -10(2/5) = \boxed{-4 < x}$

(b) $x + 3 < 2x + 8 < 3x + 10$

$x+3 < 2x+8$

$\Rightarrow 3-8 = -5 < 2x - x = x$

$-5 < x \Rightarrow x+3 < 2x+8$

$2x+8 < 3x+10$

$\Rightarrow 8-10 = -2 < 3x - 2x = x$

$-5 < \boxed{-2 < x} \Rightarrow x+3 < 2x+8 < 3x+10$

(c) $|2x-5| \leq 11 \Leftrightarrow -11 < 2x-5 < 11$

$\Rightarrow 0 < 2x-5+11 < 11+11$

$\Rightarrow 0 < 2x+6 < 22$

$\Rightarrow 0 < 2x < 16$

$\Rightarrow 0 < x < 8$ (or $x \in (0,8)$).